



Acids and bases

Acids and bases

Properties of Acids

Sour taste, Change color of dyes, Conduct electricity in solution, React with many metals, React with bases to form salts

Acids and bases

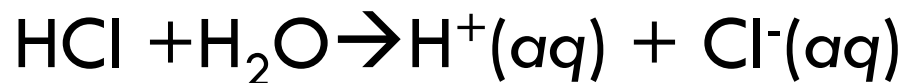
Properties of Bases

Bitter taste, Feel slippery,
Change color of dyes,
Conduct electricity in
solution, React with acids
to form salts

Acids and bases

Arrhenius definition

Arrhenius suggested that acids are compounds that contain hydrogen and can dissolve in water to release hydrogen ions into solution. For example, hydrochloric acid (HCl) dissolves in water as follows:



Acids and bases

Arrhenius definition

- Arrhenius defined bases as substances that dissolve in water to release hydroxide ions (OH^-) into solution. For example, a typical base according to the Arrhenius definition is sodium hydroxide (NaOH):
- $\text{NaOH} + \text{H}_2\text{O} \rightarrow \text{Na}^+(\text{aq}) + \text{OH}^-(\text{aq})$

Acids and bases

Arrhenius definition

- Acids release H^+ into solution and bases release OH^- . If we were to mix an acid and base together, the H^+ ion would combine with the OH^- ion to make the molecule H_2O , or plain water:
- $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}$

Acids and Bases

Bronsted Lowry theory

- The Bronsted -Lowry definition of acids is very similar to the Arrhenius definition, any substance that can donate a hydrogen ion is an acid (under the Bronsted definition, acids are often referred to as proton donors because an H^+ ion, hydrogen minus its electron, is simply a proton).

Acids and Bases

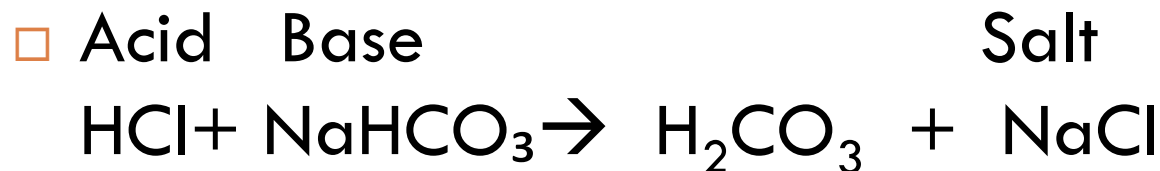
Bronsted Lowry theory

- The Bronsted definition of bases is, however, quite different from the Arrhenius definition. The Bronsted base is defined as any substance that can accept a hydrogen ion. A base is the opposite of an acid. NaOH and KOH, would still be considered bases because they can accept an H^+ from an acid to form water.

Acids and Bases

Bronsted Lowry theory

- The Bronsted-Lowry definition also explains why substances that do not contain OH^- can act like bases. Baking soda (NaHCO_3), for example, acts like a base by accepting a hydrogen ion from an acid as illustrated below:



Acids and Bases

Bronsted Lowry theory

- Any molecule or ion that can transfer a proton; a hydrogen atom nucleus to another species.
- Let us take the example of sulfuric acid
- $\text{H}_2\text{SO}_{4(l)} + \text{H}_2\text{O(l)} \rightarrow \text{H}_3\text{O}^+ (\text{aq}) + \text{HSO}_4^- (\text{aq})$
- The hydrogen sulfate ion, is also a Bronsted Lowry acid because it too can donate a proton to a water molecule.
- $\text{HSO}_4^- (\text{aq}) + \text{H}_2\text{O(l)} \rightleftharpoons \text{H}_3\text{O}^+ (\text{aq}) + \text{SO}_4^{2-} (\text{aq})$

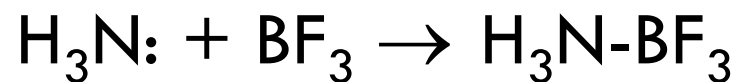
Acids and Bases

Bronsted Lowry theory

- Any atom, or molecule that receives a proton from another species.
- Ammonia is a typical base
- $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$
- In the gas phase ammonia accepts a proton from HCl and forms ammonium chloride which is composed of ammonium and chloride ion.

Lewis Acid/Base

- Acid: Electron acceptor
- Base: Electron donor



Base Acid Neutral

Species that are both acids and bases.

- Water can act as an acid by donating a proton, and can act as a base by accepting a proton.
- $\text{H}_2\text{O}(\text{l}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{OH}^- (\text{aq}) + \text{H}_3\text{O}^+ (\text{aq})$
- A species that can act as either an acid or a base is called as **amphoteric**.
- Hydrogen carbonate ion is also an example of amphoteric species.
- $\text{HCO}_3^- (\text{aq}) + \text{OH}^- (\text{aq}) \rightleftharpoons \text{CO}_3^{2-} (\text{aq}) + \text{H}_2\text{O}(\text{l})$
- It behaves as a base in the presence of an acid such as formic acid
- $\text{HCOOH} (\text{aq}) + \text{HCO}_3^- (\text{aq}) \rightleftharpoons \text{HCOO}^- (\text{aq}) + \text{H}_2\text{CO}_3(\text{aq})$

Neutralization reaction

- We have seen from the equations, acids release H^+ into solution and bases release OH^- . If we were to mix an acid and base together, the H^+ ion would combine with the OH^- ion to make the molecule H_2O , or plain water:
- $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}$. The neutralization reaction of an acid with a base will always produce water and a salt:
- | Acid | Base | Water | Salt |
|--------------|------|---------------|--|
| HCl | $+$ | NaOH | $\rightarrow \text{H}_2\text{O} + \text{NaCl}$ |
| HBr | $+$ | KOH | $\rightarrow \text{H}_2\text{O} + \text{KBr}$ |

Conjugate acids and bases

- Conjugate acids are formed when a base accepts a proton.
- Conjugate base are formed when an acid donates a proton.
- $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$
- Ammonium ion is the conjugate acid and OH ion is the conjugate base.

Strong Acids and Strong Bases

Strong acids and bases are completely ionized when dissolve in water.

Common Strong Acids	Common Strong Bases
Perchloric acid HClO_4	All Gr I metal hydroxides
Sulfuric acid H_2SO_4	Sodium Hydroxide NaOH
Nitric acid HNO_3	Potassium Hydroxide KOH etc
Hydrochloric acid HCl	All Gr II metal hydroxides
Hydrobromic acid HBr	Calcium Hydroxide Ca(OH)_2
Hydroiodic acid HI	Magnesium Hydroxide Mg(OH)_2
	Barium Hydroxide Ba(OH)_2 etc

Weak acids and bases

Weak acids and bases are partially ionized in their solutions,

Common Weak Acids	Common Weak Bases
Formic acid HCOOH	ammonia NH_3
Acetic acid CH_3COOH	Trimethyl ammonia $\text{N}(\text{CH}_3)_3$
Trichloro acetic acid CCl_3COOH	pyridine $\text{C}_5\text{H}_5\text{N}$
Hydrofluoric acid HF	Ammonium hydroxide NH_4OH
Hydrocyanic acid HCN	water H_2O
Hydrogen sulfide H_2S	
Water H_2O	

Monoprotic , diprotic, triprotic and polyprotic acids

- Monoprotic acids are acids that can donate one hydrogen ion per molecule. Example nitric acid HNO_3 or hydrochloric acid HCl .
- Diprotic acids are acids that can donate two hydrogen ions per molecule. Examples sulfuric acids H_2SO_4 .
- Triprotic acids are acids that can donate three hydrogen ions per molecule. Example phosphoric acid H_3PO_4
- Polyprotic acids are acids that can donate more than three hydrogen ions per molecule. Example- silicic acid H_4SiO_4

Self ionization of water

- Water molecules collide with one another to cause the self-ionization reaction represented by this equation: $2\text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{OH}^-$
- It is a reversible reaction so the equation is usually written with the arrows going in both directions:
$$2\text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{OH}^-(\text{aq})$$
- The self ionization of pure water produces equal amounts of H_3O^+ ions and OH^- ions. Therefore the concentrations of these ions in pure water must be equal. $[\text{OH}^-] = [\text{H}_3\text{O}^+]$

Self ionization of water

- It has been found that at 25°C the concentration of these two ions are equal to 1.00×10^{-7} mol/L
- Recall the equation for equilibrium constant

K_c which can be calculated from the equation: $2\text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{OH}^-(\text{aq})$

$$K_c = \frac{[\text{H}_3\text{O}^+][\text{OH}^-]}{[\text{H}_2\text{O}]^2}$$

The concentration of ions is extremely small, so the concentration of H_2O remains essentially constant. This gives:

$$[\text{H}_2\text{O}]^2 K_c = [\text{H}_3\text{O}^+][\text{OH}^-]$$

constant

this equilibrium constant called K_w , can be calculated as follows.

$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = (1.00 \times 10^{-7})(1.00 \times 10^{-7}) = 1.00 \times 10^{-14}$$

K_w is called as auto ionisation constant.

Acid ionization constant

- The equilibrium constant of the ionization of a weak acid in water is known as the acid ionization constant K_a .
- $\text{CH}_3\text{COOH}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{CH}_3\text{COO}^- (\text{aq}) + \text{H}_3\text{O}^+ (\text{aq})$
- The equilibrium expression for this reaction is written as follows
- $K_a = [\text{H}_3\text{O}^+][\text{CH}_3\text{COO}^-] / [\text{CH}_3\text{COOH}] = 1.75 \times 10^{-5}$

pH

- What is pH?
- The negative logarithm of the hydronium ion concentration in a solution.
- pH of a solution ranging from 1-7 are usually acidic.
- pH of a solution ranging from 7-14 are usually basic.
- pH of a solution which is 7 are neutral.

pH

- In a **neutral solution**, the concentrations of $\text{H}^+(\text{aq})$ and $\text{OH}^-(\text{aq})$ are **equal**, as they are in pure water.
- In an **acidic** solution, the concentration of $\text{H}^+(\text{aq})$ is greater than that of $\text{OH}^-(\text{aq})$.
- In a **basic** solution, the concentration of $\text{OH}^-(\text{aq})$ is greater than that of $\text{H}^+(\text{aq})$.
- In an acidic solution, $[\text{H}^+] > 1.0 \times 10^{-7} \text{ M}$.
- In a neutral solution, $[\text{H}^+] = 1.0 \times 10^{-7} \text{ M}$.
- In a basic solution, $[\text{H}^+] < 1.0 \times 10^{-7} \text{ M}$.

pH

A sample of orange juice has a hydrogen-ion concentration of 2.9×10^{-4} M. What is the pH?

$$\text{pH} = -\log[\text{H}^+]$$

$$\text{pH} = -\log(2.9 \times 10^{-4})$$

$$\text{pH} = 3.54$$

pH

Example : An ammonia solution has a hydroxide-ion concentration of 1.9×10^{-3} M. What is the pH of the solution?

You first calculate the pOH:

$$\text{pOH} = -\log(1.9 \times 10^{-3}) = 2.72$$

Then the pH is:

$$\text{pH} = 14.00 - 2.72 = 11.28$$

Buffers

- A buffer solution is one which resists changes in pH when small quantities of an acid or an alkali are added to it. Solutions of a weak acid and its conjugate base or weak base and its conjugate acid are able to maintain pH and are buffer solutions
- Buffer solutions are used in fermentation, food preservatives, drug delivery, electroplating, printing and the activity of enzymes, and the blood oxygen-carrying capacity needs specific hydrogen ion concentration (pH).

Types of Buffer Solutions

□ **Acidic Buffers**

- As the name suggests, these solutions are used to maintain acidic environments. Acid buffer has acidic pH and is prepared by mixing a weak acid and its salt with a strong base. An aqueous solution of an equal concentration of acetic acid and sodium acetate has a pH of 4.74.
- The pH of these solutions is below seven.
- These solutions consist of a weak acid and a salt of a weak acid.
- An example of an acidic buffer solution is a mixture of sodium acetate and acetic acid ($\text{pH} = 4.75$).

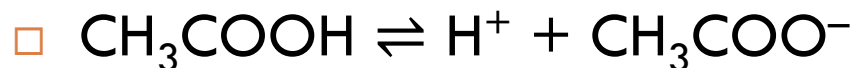
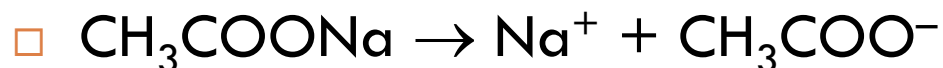
Types of Buffer Solutions

□ Alkaline Buffers

- These buffer solutions are used to maintain basic conditions. A basic buffer has a basic pH and is prepared by mixing a weak base and its salt with strong acid. The aqueous solution of an equal concentration of ammonium hydroxide and ammonium chloride has a pH of 9.25.
- The pH of these solutions is above seven.
- They contain a weak base and a salt of the weak base.
- An example of an alkaline buffer solution is a mixture of ammonium hydroxide and ammonium chloride ($\text{pH} = 9.25$).

Mechanism of a Buffering Action

In solution, the salt is completely ionised, and the weak acid is partly ionised.

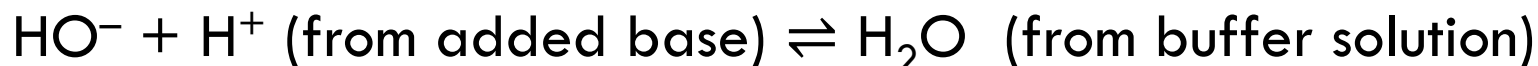


On Addition of Acid and Base

1. On addition of acid, the released protons of acid will be removed by the acetate ions to form an acetic acid molecule.



2. On addition of the base, the hydroxide released by the base will be removed by the hydrogen ions to form water.



Buffer solution related Problems

Problem 1: What is the pH of a buffered solution of 0.5 M ammonia and 0.5 M ammonium chloride when enough hydrochloric acid corresponds to make 0.15 M HCl?

Sol: The pK_b of ammonia is 4.75; $pK_a = 14 - pK_b = 9.25$

- 0.15 M H^+ reacts with 0.15 M ammonia to form 0.15 M more ammonium. So, the ammonium ion is 0.65 M and 0.35 M remaining ammonia (base).
- Using the Henderson-Hasselbalch equation,
- $pK_a - \log ([\text{salt}]/[\text{acid}]) = 9.25 - \log (.65/.35) = 9.25 - .269 = 8.98$

Buffer solution related Problems

Problem 2: How many moles of sodium acetate and acetic acid must you use to prepare 1.00 L of a 0.100 mol/L buffer with pH 5.00? [pKa of acetic acid is 4.74]

Sol: $\text{pH} = \text{pKa} + \log([A^-]/[HA])$

$$5.00 = 4.74 + \log([A^-]/[HA]); \log([A^-]/[HA]) = 0.26$$

$$[A^-]/[HA] = 10^{0.26} = 1.82$$

$$[A^-] = 1.82[HA]$$

$$\text{Also, } [A^-] + [HA] = 0.100 \text{ mol/L; } 1.82[HA] + [HA] = 0.100 \text{ mol/L}$$

$$2.82[HA] = 0.100 \text{ mol/L}$$

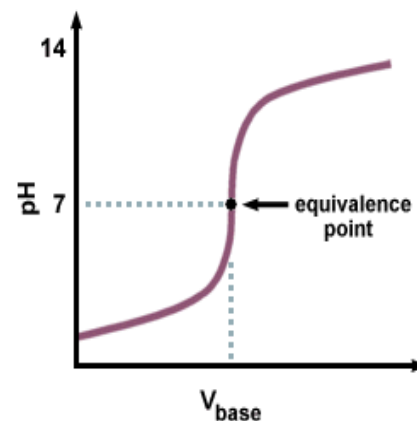
$$[HA] = 0.0355 \text{ mol/L}$$

$$[A^-] = (0.100 - 0.0355) \text{ mol/L} = 0.0645 \text{ mol/L}$$

0.0355 mol of acetic acid and 0.0645 mol of sodium acetate is required to prepare 1 L of the buffer solution.

Titration

- A method of determining the concentration of one solution by reacting it with a standard solution. This can be done by the operation of gradually adding one solution to another to reach an equivalence point.



Titration

- Equivalence point is the point in a titration when the amount of added base or acid exactly equals the amount of acid or base originally in solution.
- Titrant: The solution added to another solution in a titration.
- Indicator: An organic dye that changes color when the pH changes.